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Machine Learning Systems Engineer, Research Tools

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📍 San Francisco, CA | New York City, NY | Seattle, WA

About Anthropic

Anthropic's mission is to create reliable, interpretable, and steerable AI systems. We want AI to be safe and beneficial for our users and for society as a whole. Our team is a quickly growing group of committed researchers, engineers, policy experts, and business leaders working together to build beneficial AI systems.

About the Role:

We are seeking an experienced Machine Learning Systems Engineer to join our Encodings and Tokenization team at Anthropic. This cross-functional role will be instrumental in developing and optimizing the encodings and tokenization systems used throughout our Finetuning workflows. As a bridge between our Pretraining and Finetuning teams, you'll build critical infrastructure that directly impacts how our models learn from and interpret data. Your work will be foundational to Anthropic's research progress, enabling more efficient and effective training of our AI systems while ensuring they remain reliable, interpretable, and steerable.

Responsibilities:

- Design, develop, and maintain tokenization systems used across Pretraining and Finetuning workflows
- Optimize encoding techniques to improve model training efficiency and performance
- Collaborate closely with research teams to understand their evolving needs around data representation
- Build infrastructure that enables researchers to experiment with novel tokenization approaches
- Implement systems for monitoring and debugging tokenization-related issues in the model training pipeline
- Create robust testing frameworks to validate tokenization systems across diverse languages and data types
- Identify and address bottlenecks in data processing pipelines related to tokenization
- Document systems thoroughly and communicate technical decisions clearly to stakeholders across teams

You May Be a Good Fit If You:

- Have significant software engineering experience with demonstrated machine learning expertise



- Are comfortable navigating ambiguity and developing solutions in rapidly evolving research environments
- Can work independently while maintaining strong collaboration with cross-functional teams
- Are results-oriented, with a bias towards flexibility and impact
- Have experience with machine learning systems, data pipelines, or ML infrastructure
- Are proficient in Python and familiar with modern ML development practices
- Have strong analytical skills and can evaluate the impact of engineering changes on research outcomes
- Pick up slack, even if it goes outside your job description
- Enjoy pair programming (we love to pair!)
- Care about the societal impacts of your work and are committed to developing AI responsibly

Strong Candidates May Also Have Experience With:

- Working with machine learning data processing pipelines
- Building or optimizing data encodings for ML applications
- Implementing or working with BPE, WordPiece, or other tokenization algorithms
- Performance optimization of ML data processing systems
- Multi-language tokenization challenges and solutions
- Research environments where engineering directly enables scientific progress
- Distributed systems and parallel computing for ML workflows
- Large language models or other transformer-based architectures (not required)

Deadline to apply: None. Applications will be reviewed on a rolling basis.

The annual compensation range for this role is listed below.

For sales roles, the range provided is the role's On Target Earnings ("OTE") range, meaning that the range includes both the sales commissions/sales bonuses target and annual base salary for the role.

Annual Salary:

\$320,000 - \$405,000 USD

Logistics

Minimum education: Bachelor's degree or an equivalent combination of education, training, and/or experience

Required field of study: A field relevant to the role as demonstrated through coursework, training, or professional experience

Minimum years of experience: Years of experience required will correlate with the internal job level requirements for the position

Location-based hybrid policy: Currently, we expect all staff to be in one of our offices at least 25% of the time. However, some roles may require more time in our offices.

Visa sponsorship: We do sponsor visas! However, we aren't able to successfully sponsor visas for every role and every candidate. But if we make you an offer, we will make every reasonable effort to get you a visa, and we retain an immigration lawyer to help with this.

We encourage you to apply even if you do not believe you meet every single qualification. Not all strong candidates will meet every single qualification as listed. Research shows that people who identify as being from underrepresented groups are more prone to experiencing imposter syndrome and doubting the strength of their candidacy, so we urge you not to exclude yourself prematurely and to submit an application if you're interested in this work. We think AI systems like the ones we're building have enormous social and ethical implications. We think this makes representation even more important, and we strive to include a range of diverse perspectives on our team.

Your safety matters to us. To protect yourself from potential scams, remember that Anthropic recruiters only contact you from @anthropic.com email addresses. In some cases, we may partner with vetted recruiting agencies who will identify themselves as working on behalf of Anthropic. Be cautious of emails from other domains. Legitimate Anthropic recruiters will never ask for money, fees, or banking information before your first day. If you're ever unsure about a communication, don't click any links—visit anthropic.com/careers directly for confirmed position openings.

How we're different

We believe that the highest-impact AI research will be big science. At Anthropic we work as a single cohesive team on just a few large-scale research efforts. And we value impact — advancing our long-term goals of steerable, trustworthy AI — rather than work on smaller and more specific puzzles. We view AI research as an empirical science, which has as much in common with physics and biology as with traditional efforts in computer science. We're an extremely collaborative group, and we host frequent research discussions to ensure that we are pursuing the highest-impact work at any given time. As such, we greatly value communication skills.

The easiest way to understand our research directions is to read our recent research. This research continues many of the directions our team worked on prior to Anthropic, including: GPT-3, Circuit-Based Interpretability, Multimodal Neurons, Scaling Laws, AI & Compute, Concrete Problems in AI Safety, and Learning from Human Preferences.

Come work with us!

Anthropic is a public benefit corporation headquartered in San Francisco. We offer competitive compensation and benefits, optional equity donation matching, generous vacation and parental leave, flexible working hours, and a lovely office space in which to collaborate with colleagues. **Guidance on Candidates' AI Usage:** Learn about [our policy](#) for using AI in our application process.



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Interested in building your career at Anthropic? Get future opportunities sent straight to your email.

[Create alert](#)

Apply for this job

* indicates a required field

First Name *

Last Name *

Email *

Country ▼

Phone

Resume/CV

Attach

Dropbox

Google Drive

Enter manually

Accepted file types: pdf, doc, docx, txt, rtf

(Optional) Personal Preferences *

How do you pronounce your name?

LinkedIn Profile

Please ensure to provide *either* your LinkedIn profile or Resume, we require at least one of the two.

Website

Publications (e.g. Google Scholar) URL

GitHub URL

Are you open to working in-person in one of our offices 25% of the time? *

Select...



When is the earliest you would want to start working with us?

Do you have any deadlines or timeline considerations we should be aware of?

AI Policy for Application *

Select...



We believe that AI will have a transformative impact on the world, and we're seeking exceptional candidates who collaborate thoughtfully with Claude to realize this vision. At the same time, we want to understand your unique skills, expertise, and perspective through our hiring process. We invite you to review our [AI partnership guidelines for candidates](#) and confirm your understanding by selecting "Yes."

Why Anthropic? *

Why do you want to work at Anthropic? (We value this response highly - great answers are often 200-400 words.)

Do you require visa sponsorship? *

Select...



Will you now or will you in the future require employment visa sponsorship to work in the country in which the job you're applying for is located? *

Select...



Additional Information *

Add a cover letter or anything else you want to share.

Are you open to relocation for this role? *

Select...

What is the address from which you plan on working? If you would need to relocate, please type "relocating".

Have you ever interviewed at Anthropic before? *

Select...

Voluntary Self-Identification

For government reporting purposes, we ask candidates to respond to the below self-identification survey. Completion of the form is entirely voluntary. Whatever your decision, it will not be considered in the hiring process or thereafter. Any information that you do provide will be recorded and maintained in a confidential file.

As set forth in Anthropic's Equal Employment Opportunity policy, we do not discriminate on the basis of any protected group status under any applicable law.

Gender

Select...

Are you Hispanic/Latino?

Select...

Race & Ethnicity Definitions

If you believe you belong to any of the categories of protected veterans listed below, please indicate by making the appropriate selection. As a government contractor subject to the Vietnam Era Veterans Readjustment Assistance Act (VEVRAA), we request this information in order to measure the effectiveness of the outreach and positive recruitment efforts we undertake pursuant to VEVRAA. Classification of protected categories is as follows:

A "disabled veteran" is one of the following: a veteran of the U.S. military, ground, naval or air service who is entitled to compensation (or who but for the receipt of military retired pay would be entitled to compensation)

under laws administered by the Secretary of Veterans Affairs; or a person who was discharged or released from active duty because of a service-connected disability.

A "recently separated veteran" means any veteran during the three-year period beginning on the date of such veteran's discharge or release from active duty in the U.S. military, ground, naval, or air service.

An "active duty wartime or campaign badge veteran" means a veteran who served on active duty in the U.S. military, ground, naval or air service during a war, or in a campaign or expedition for which a campaign badge has been authorized under the laws administered by the Department of Defense.

An "Armed forces service medal veteran" means a veteran who, while serving on active duty in the U.S. military, ground, naval or air service, participated in a United States military operation for which an Armed Forces service medal was awarded pursuant to Executive Order 12985.

Veteran Status

Select...

Voluntary Self-Identification of Disability

Form CC-305

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OMB Control Number 1250-0005

Expires 04/30/2026

Why are you being asked to complete this form?

We are a federal contractor or subcontractor. The law requires us to provide equal employment opportunity to qualified people with disabilities. We have a goal of having at least 7% of our workers as people with disabilities. The law says we must measure our progress towards this goal. To do this, we must ask applicants and employees if they have a disability or have ever had one. People can become disabled, so we need to ask this question at least every five years.

Completing this form is voluntary, and we hope that you will choose to do so. Your answer is confidential. No one who makes hiring decisions will see it. Your decision to complete the form and your answer will not harm you in any way. If you want to learn more about the law or this form, visit the U.S. Department of Labor's Office of Federal Contract Compliance Programs (OFCCP) website at www.dol.gov/ofccp.

How do you know if you have a disability?

A disability is a condition that substantially limits one or more of your "major life activities." If you have or have ever had such a condition, you are a person with a disability. **Disabilities include, but are not limited to:**

- Alcohol or other substance use disorder (not currently using drugs illegally)
- Autoimmune disorder, for example, lupus, fibromyalgia, rheumatoid arthritis, HIV/AIDS
- Blind or low vision
- Cancer (past or present)
- Cardiovascular or heart disease

- Celiac disease
- Cerebral palsy
- Deaf or serious difficulty hearing
- Diabetes
- Disfigurement, for example, disfigurement caused by burns, wounds, accidents, or congenital disorders
- Epilepsy or other seizure disorder
- Gastrointestinal disorders, for example, Crohn's Disease, irritable bowel syndrome
- Intellectual or developmental disability
- Mental health conditions, for example, depression, bipolar disorder, anxiety disorder, schizophrenia, PTSD
- Missing limbs or partially missing limbs
- Mobility impairment, benefiting from the use of a wheelchair, scooter, walker, leg brace(s) and/or other supports
- Nervous system condition, for example, migraine headaches, Parkinson's disease, multiple sclerosis (MS)
- Neurodivergence, for example, attention-deficit/hyperactivity disorder (ADHD), autism spectrum disorder, dyslexia, dyspraxia, other learning disabilities
- Partial or complete paralysis (any cause)
- Pulmonary or respiratory conditions, for example, tuberculosis, asthma, emphysema
- Short stature (dwarfism)
- Traumatic brain injury

Disability Status

Select... ▼

PUBLIC BURDEN STATEMENT: According to the Paperwork Reduction Act of 1995 no persons are required to respond to a collection of information unless such collection displays a valid OMB control number. This survey should take about 5 minutes to complete.

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